

2ND YEAR

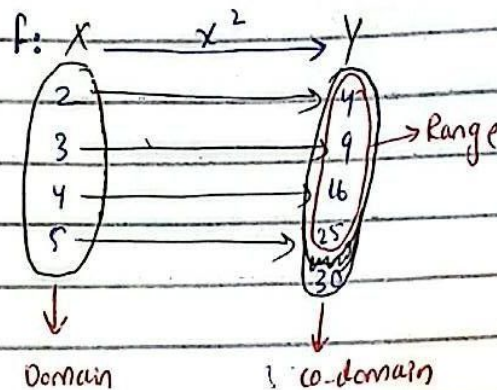


Chapter # 01

Functions, Limits & Continuity

i- A function "f" from a non-empty set X to Y is a relation, such that each element of X is related to a unique element of Y.

$$f: X \rightarrow Y$$



$$f = \{(2, 4), (3, 9), (4, 16), (5, 25)\}$$

$$(2, 4) \mid (2, 8)$$

2 $\begin{matrix} \nearrow 4 \\ \searrow 8 \end{matrix}$ uniqueness ends.

ii- $y = x^2$
 $\downarrow$
 Range
 Repeats

$$y^2 = x + 1 \quad (\text{Not a function})$$

jis relationship mai y pr square a jaye wo function nahi hota.

$$y^2 = x + 1$$

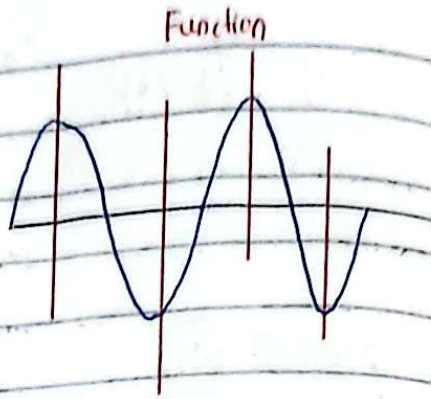
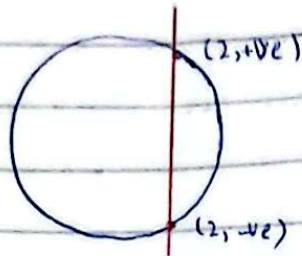
$$x = 8$$

$$y^2 = 9, |y| = \pm 3$$

$y^2 \Rightarrow$ domain repeats

$$(8, 3), (8, -3)$$

iii - Vertical line test:

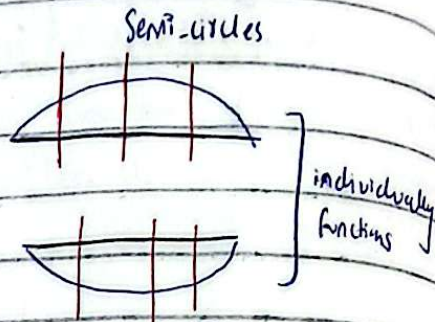


$$x^2 + y^2 = 9 \Rightarrow y = \pm \sqrt{9 - x^2}$$

causing factor

$$y = +\sqrt{9 - x^2}$$

$$y = -\sqrt{9 - x^2}$$



Types of Function:

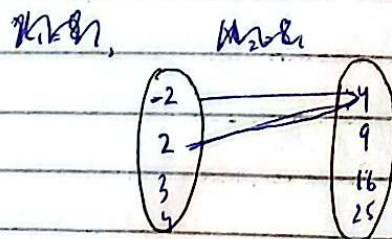
(i) On-to Function: ^(Surjective Function) if $\text{Range}(f) = Y = \text{co-domain}$.

(ii) In-to Function: if $\text{Range}(f) \subset Y$

(iii) One-one Function: ^(Injective Function) if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

(OR)

if $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$



$$\begin{aligned} -2 = x_1 & ; x_2 = 2 \\ f(-2) &= f(2) \\ 4 &= 4 \\ -2 \neq 2 & \text{ But } 4 = 4 \end{aligned} \quad \left. \vphantom{\begin{aligned} -2 = x_1 & ; x_2 = 2 \\ f(-2) &= f(2) \\ 4 &= 4 \\ -2 \neq 2 & \text{ But } 4 = 4 \end{aligned}} \right\} \text{violation.}$$

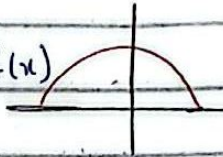
$$f(x) = x^2 ; x \in \mathbb{R} \rightarrow \text{not one-one}$$

$$g(x) = x^2 ; x \in \mathbb{R}^+ \rightarrow \text{one-one}$$

(iv) Bijective Function (one-to-one function)

if f is both one-one & onto

Even Function :- if $f(-x) = f(x)$



Range ka pata na ho
hum use onto man
lete hain

odd Function :- if $f(-x) = -f(x)$



Every even function is symmetric about y-axis.

Every odd function is symmetric about origin.

Explicit function/Relation :- y ko alag kar sakte hain

$$xy + y - 10 = 0$$

$$y(x+1) = 10$$

$$y = \frac{10}{x+1}$$

$$y = f(x)$$

Implicit function/Relation :- y ko alag na kar sakte hain.

$$xy^2 + y - 10 = 0$$

$$y \neq f(x)$$

$$\text{So, } f(x, y) = 0$$

if y power in whole function is same $\Rightarrow$ Explicit

if y power is different $\Rightarrow$ Implicit

are like trigonometric functions.

$e = 2.71 \rightarrow$ upto 2 answers

$$\frac{e^0 - e^0}{2}$$

Hyperbolic Functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

Domain: $\mathbb{R}$

Range: $\mathbb{R}$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Domain: $\mathbb{R}$

Range: $[1, +\infty)$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

As $e^x > 0$ $\frac{e^x}{e^x} \neq 0$
exponential functions always
gives answer greater than zero.

$$e^{-(-5)} = e^5$$

$$e^{-5} = \frac{1}{e^5}$$

$$e^8 > \frac{1}{e^5}$$

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

multiplication

$$\sin(2x) = 2 \sin x \cos x$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

division formula don't change

$$\sinh(2x) = 2 \sinh x \cosh x$$

$$\tanh(x) = \frac{\sinh x}{\cosh x}$$

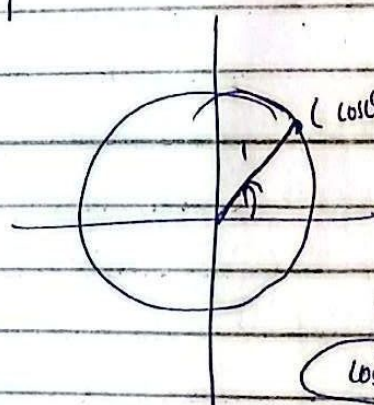
$$\cosh^2 x - \sinh^2 x = 1$$

↓

1- Divide by $\sin$

2- Divide by $\cos$

$\sin, \cos$
has link with circle



any point

every point makes $\sin$ & $\cos$

$$x^2 + y^2 = 1$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

hyperbola

$$x^2 - y^2 = 1$$

$$\cosh^2 x - \sinh^2 x = 1$$

hyperbolic function

satisfy identity of hyperbolic functions.

Exponential Function:

$$f(x) = a^x ; a > 0, a \neq 1$$

$$\text{Domain} = \mathbb{R}$$

$$\text{Range} = (0, +\infty)$$

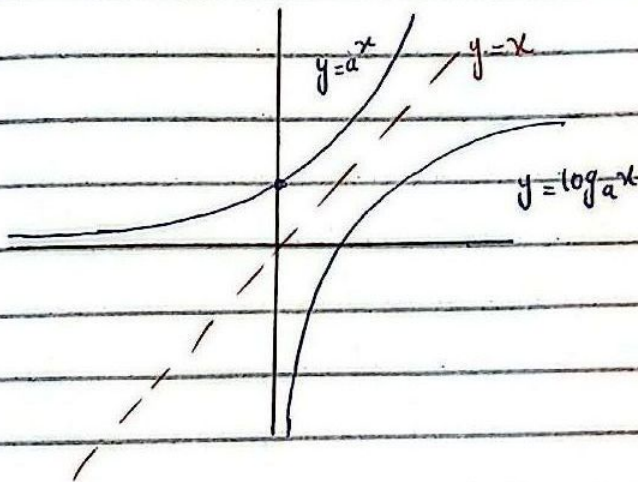
Logarithmic Function:

$$f(x) = \log_a x ; a > 0, a \neq 1$$

$$\text{Domain} = (0, +\infty)$$

$$\text{Range} = \mathbb{R}$$

They both are inverse of each other.



Limits of a function :-

CASE (i) $\lim_{x \rightarrow a} f(x) = \text{unique real number.}$

CASE (ii) $\lim_{x \rightarrow a} f(x) = \pm \infty$ (limit does not exist)

* CASE (iii) $\lim_{x \rightarrow a} f(x) = \left(\frac{0}{0}\right) \text{ OR } \left(\frac{\infty}{\infty}\right)$ "Indeterminate form"

$$(i) \lim_{x \rightarrow 0} \frac{1 + \sin x}{x + 1} = \frac{1 + \sin(0)}{0 + 1} = \frac{1}{1} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad (or) \quad \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$$

$$\lim_{x \rightarrow 0} \frac{3x}{\sin 3x} = \frac{1}{3} (1) = \frac{1}{3}$$

$$(ii) \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = ? \quad -1 \leq \sin x \leq 1$$

A (Any value) ∞ Any no. divided by ∞ gives answer 0

= 0

$$\lim_{x \rightarrow 0} \frac{|x|}{x} = ? \quad \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -x = -1$$

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \frac{+x}{x} = +1$$

Absolute value Function:

$$|x| = \begin{cases} -x & , x < 0 \\ x & , x \geq 0 \end{cases}$$

L.H.L $\neq$ R.H.L
Limit does not exist

$$f(x) = \frac{1}{x}$$

$$f(x) = \frac{1}{0} \text{ (undefined) } \text{ not exact zero}$$

But:-

$$\lim_{x \rightarrow 0^-} f(x) = \frac{1}{0} = -\infty \quad (\text{not exact zero})$$

$$\lim_{x \rightarrow 0^+} f(x) = \frac{1}{0} = +\infty$$

$$f(-0.0001) = -10000$$

$$\downarrow \quad \downarrow$$

$$0 \quad -\infty$$

$$f(x) = \frac{1}{x}$$

$$f(0.0001) = \frac{1}{0.0001} = 10000$$

$$\downarrow \quad \downarrow$$

$$0 \quad +\infty$$

$$-x\sqrt{x^2} = -x|x| = -x$$

$$\lim_{x \rightarrow 0} \frac{1}{x} = \pm \infty$$

belongs to 1st category.

$$\lim_{x \rightarrow \infty} \frac{3x+2}{\sqrt{x^2+9}} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{2x+3}{3x+1} = \left(\frac{\infty}{\infty}\right)$$

$$\frac{x + (3 + 2/x)}{x \sqrt{1 + 9/x^2}} = \frac{3+0}{\sqrt{1+0}}$$

(X)

$$\frac{x(2 + 3/x)}{x(3 + 1/x)}$$

$$\frac{2+0}{3+0} = \left[\frac{2}{3}\right]$$

$$\lim_{x \rightarrow \infty} \frac{x(3 + 2/x)}{-x \sqrt{1 + 9/x^2}} = \frac{3+0}{-\sqrt{1+0}}$$

$$= (-3)$$

$$\sqrt{x^2} = |x| = \begin{cases} x & ; x > 0 \\ -x & , x < 0 \end{cases}$$

$$\lim_{x \rightarrow \infty} \left(\frac{2x+3}{3x+1} \right)$$

if degree same of numerators & denominators.
2/3 answers will be ^{leading} coefficient. of L.H.S

CASE (i): $\lim_{x \rightarrow \infty} \frac{ax^3 + bx^2 + \dots}{cx^3 + dx^2 + \dots} = a/c$

CASE (ii): $\lim_{x \rightarrow \infty} \frac{ax^3 + bx^2 + \dots}{cx^2 + dx + \dots} = \infty$

CASE (iii): $\lim_{x \rightarrow \infty} \frac{ax^2 + bx + \dots}{cx^3 + dx^2 + \dots} = 0$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} = (0/0) \text{ form}$$

$$1 - \cos x = 2 \sin^2 \frac{x}{2}$$

~~2 \sin^2 \frac{x}{2}~~

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} \times \frac{1 + \cos x}{1 + \cos x} = \frac{1 - \cos^2 x}{\sin x (1 + \cos x)} = \frac{\sin^2 x}{\sin x (1 + \cos x)}$$

$$= \frac{0}{1 + \cos(0)} = \frac{0}{2}$$

TRICK

For 0/0 question

First take derivative of both

$$(i) \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} \quad (0/0)$$

$$\frac{0 - (-\sin x)}{\cos x} = \frac{\sin x}{\cos x}$$

And now apply limit

$$\frac{\sin(0)}{\cos(0)} = 0$$

$$(ii) \lim_{x \rightarrow 2} \frac{x^2 - 4}{(x - 2)} = 0/0$$

$$\frac{2x - 0}{1 - 0} = 2x$$

$$2(2) = 4$$

$$(iii) \lim_{x \rightarrow 1} \frac{\ln x}{1 - x} = \frac{0}{1 - 1} = 0/0$$

$$\frac{\text{derivative}}{a - 1} \quad \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$-\frac{1}{1^2} = -1$$

Also works for (∞/∞) form

but we have other ~~more~~ easy methods for solving it that's why we prefer other methods.

Results:

$$(i) \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$(ii) \lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$(iii) \lim_{x \rightarrow \infty} (1+1/x)^x = e$$

$$\begin{aligned} \Rightarrow \lim_{x \rightarrow 0} (1+3x)^{1/x} &= \left[\lim_{x \rightarrow 0} (1+3x)^{1/3x} \right]^3 \\ &= [e]^3 \\ &= \boxed{e^3} \end{aligned}$$

$$\begin{aligned} \Rightarrow \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{2x} &= \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{-x} \cdot \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{-x} \\ &= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{-x}\right)^{-x} \right]^{-2} \\ &= [e]^{-2} = \frac{1}{e^2} \end{aligned}$$

multiplying & divided by minus.

Continuity of a function:

A function $f(x)$ is said to be continuous at $x=c$ if

i) $f(c)$ is defined (i.e. real number).

ii) $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$ (i.e. limit $f(x)$ exists).

iii) $f(c) = \lim_{x \rightarrow c} f(x)$

3 conditions when multiple branches.

Q#01 $f(x) = \frac{x+1}{(x-2)(x-3)}$] 1 branch so we will check only 1st point

$f(x)$ is continuous at $x=1, 2, 3, 4$

$f(x)$ is continuous at $x=1, 4$

$f(x)$ is discontinuous at $x=2, 3$

Q#02 $f(x) = \begin{cases} 3x-1 & ; x < 1 \\ x^2+1 & ; x = 1 \\ 5x-3 & ; x > 1 \end{cases}$ no. given here, at that no. func will be discontinuous
only discontinuous at 1
doubt on 1 only

at $x=1$?

Put 1 in all three, we get

$$f(1) = 2$$

$$f(1) = 2$$

$$f(1) = 2$$

Function is continuous.

Q#03 $f(x) = \begin{cases} 3x+m & ; x < 1 \\ x^2+1 & ; x = 1 \\ 5x+n & ; x > 1 \end{cases}$

Given $f(x)$ is continuous. (at all & at 1 also)

Put 1 in all branches.

$$3+m = 2 = 5+n$$

$$3+m = 2 \quad 2 = 5+n$$

$$m = -1$$

$$n = -3$$